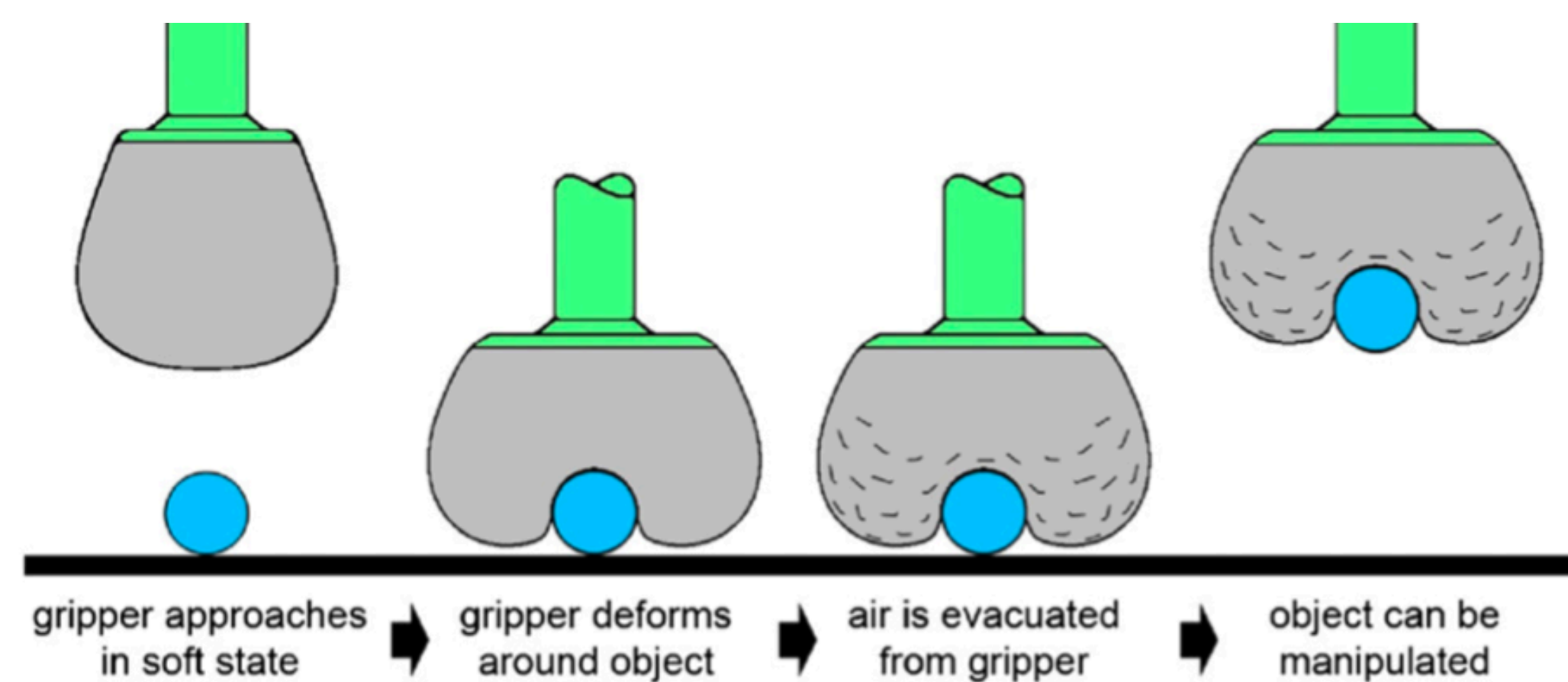


Background



By creating a mold of the object, a vacuum pump is then used to create a process known as jamming, effectively solidifying the gripper and locking the object in place.

In 2010, the Proceedings of National Sciences (PNAS) published a research article on:

- The **development of a universal granular gripper**.
 - Utilizes a vacuum pump to **induce jamming** (transition of the gripper from a fluid to a solid-state).

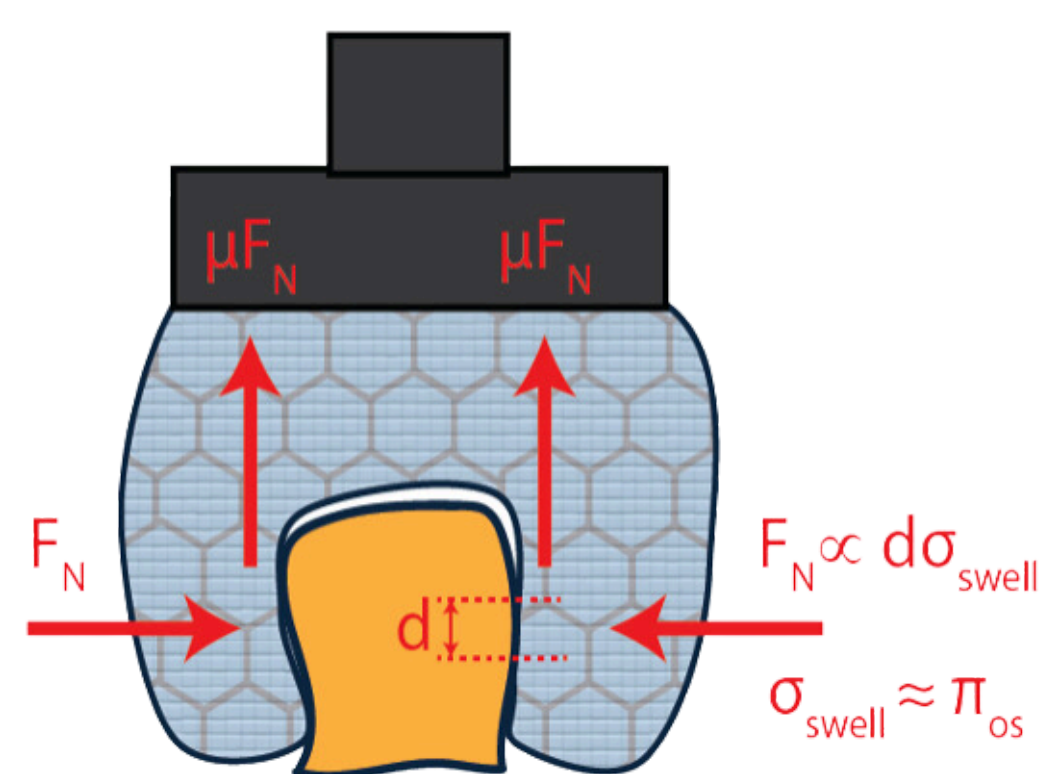
However, this gripper is severely **limited by its vacuum pump** due to:

- Bulkier gripper design** due to airflow management.
- Reliance on human operation** for actuation.
- Grasping force is **limited by the efficiency of the vacuum pump**.
- Difficulties of **navigating** a bulky gripper **within the human body**.

Our Approach

Replace vacuum-based jamming with stimuli-responsive hydrogel beads that induced jamming when swelled. The decision to use beads over a monolithic hydrogel includes:

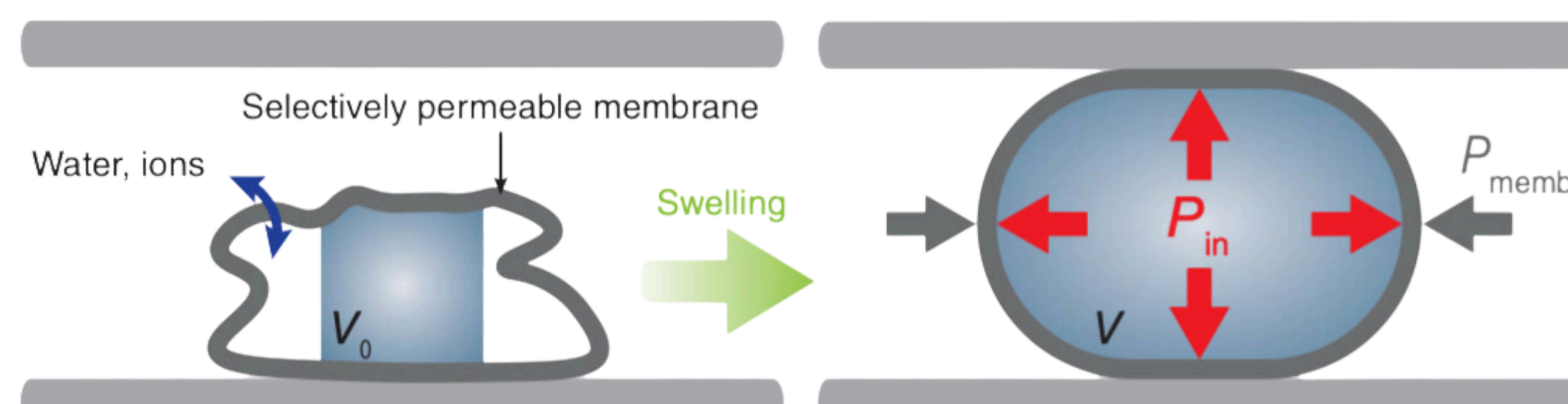
- Faster actuation times** due to faster water flow between the gaps of the beads.
- Better conformability** to objects due to bead slippage.
- Tunability of bead stiffnesses** as per swelling and deswelling processes.
- More efficient stimuli-responsive actuation** (e.g. water, temp., pH).



The initial proposal of a universal hydrogel bead-actuated gripper.

However, **hydrogels aren't known to generate high actuation forces**. Therefore, and backed by a recent study, we aim to:

- Confine our hydrogel beads in a **stiff, flexible, and porous membrane** to generate effective grasping forces.



A study in 2022 revealed that confining a swelling hydrogel in a non-expanding membrane generates high actuation forces.

Methods

Membrane Fabrication



Hydrogel bead packet (material: PETG)



3D printed gripper CAD



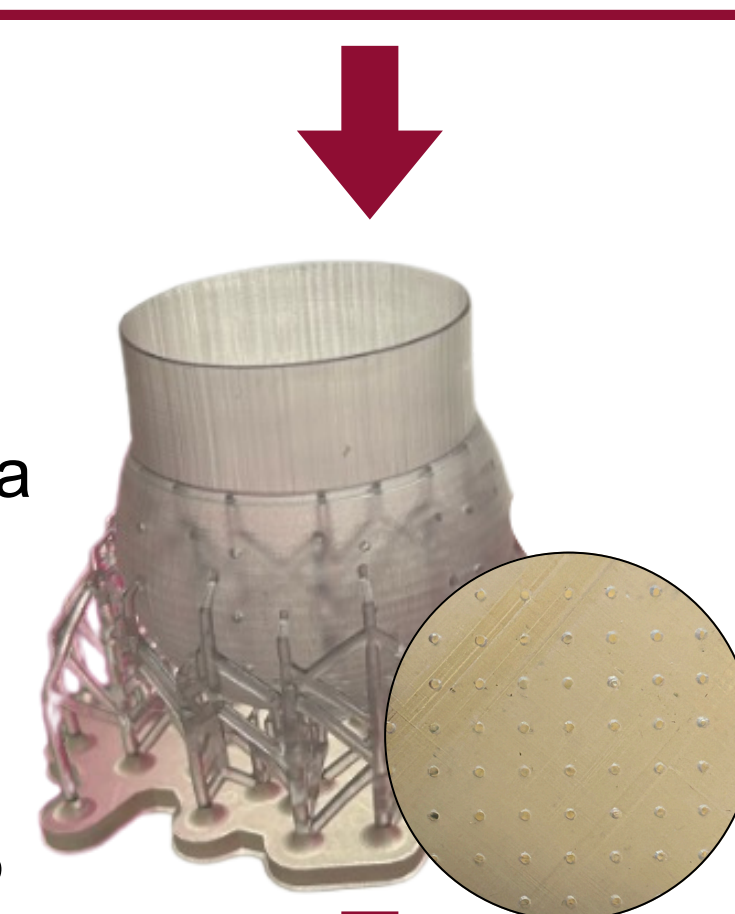
Casings for silicone rubber mold

Options of Approach :

- Joining thin **films** to create a **membrane**.
- 3D printing** the membrane using SLS/resin printing.
- Molding the membrane** from silicone rubber.

Chosen Resin Material: Flexible 80A

- Ultimate Tensile Strength:** 8.9 MPa
- Stress at 50% Elongation:** 3.1 MPa
- Stress at 100% Elongation:** 6.3 MPa
- Elongation at Break:** 120%
- Tear Strength:** 24 kN/m
- Shore Hardness:** 80A
- 24 Hour Weight Gain (Water):** 0.7%



Gripper Dimensions:

- Membrane thickness:** 0.5 mm
- Sphere diameter:** 43 mm
- Inside Collar:** 35 mm
- Outside Collar:** 35.5 mm
- Porosity Density:** 0.01 holes/mm²

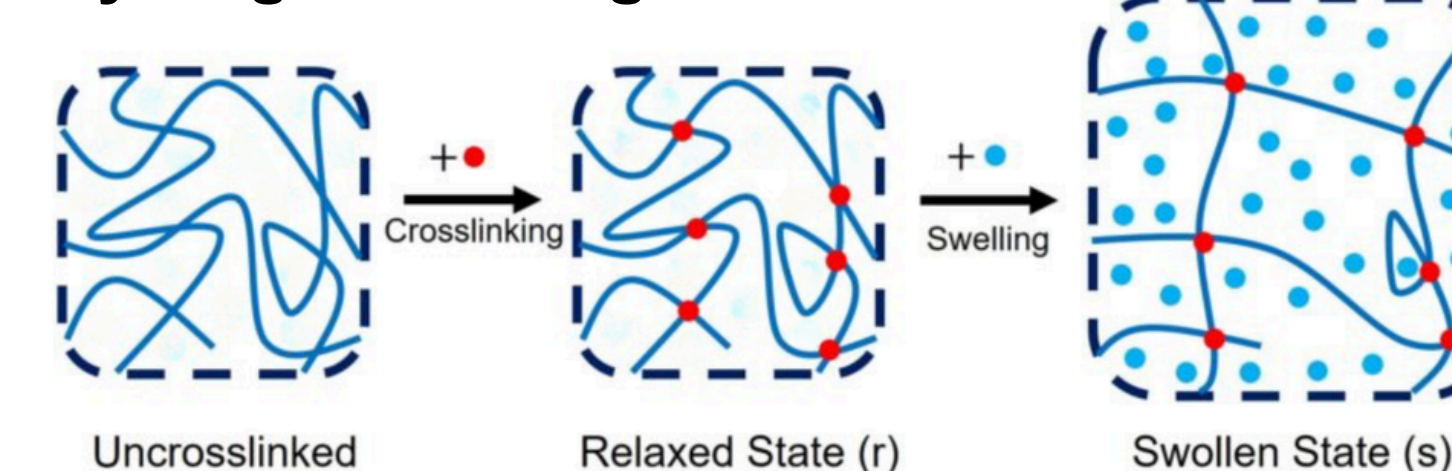
Hydrogel Beads:

- Beads were not synthesized but carefully chosen and purchased.



MagicClub Water Beads; The diagram depicts free-swelling conditions with water.

Hydrogel Swelling:

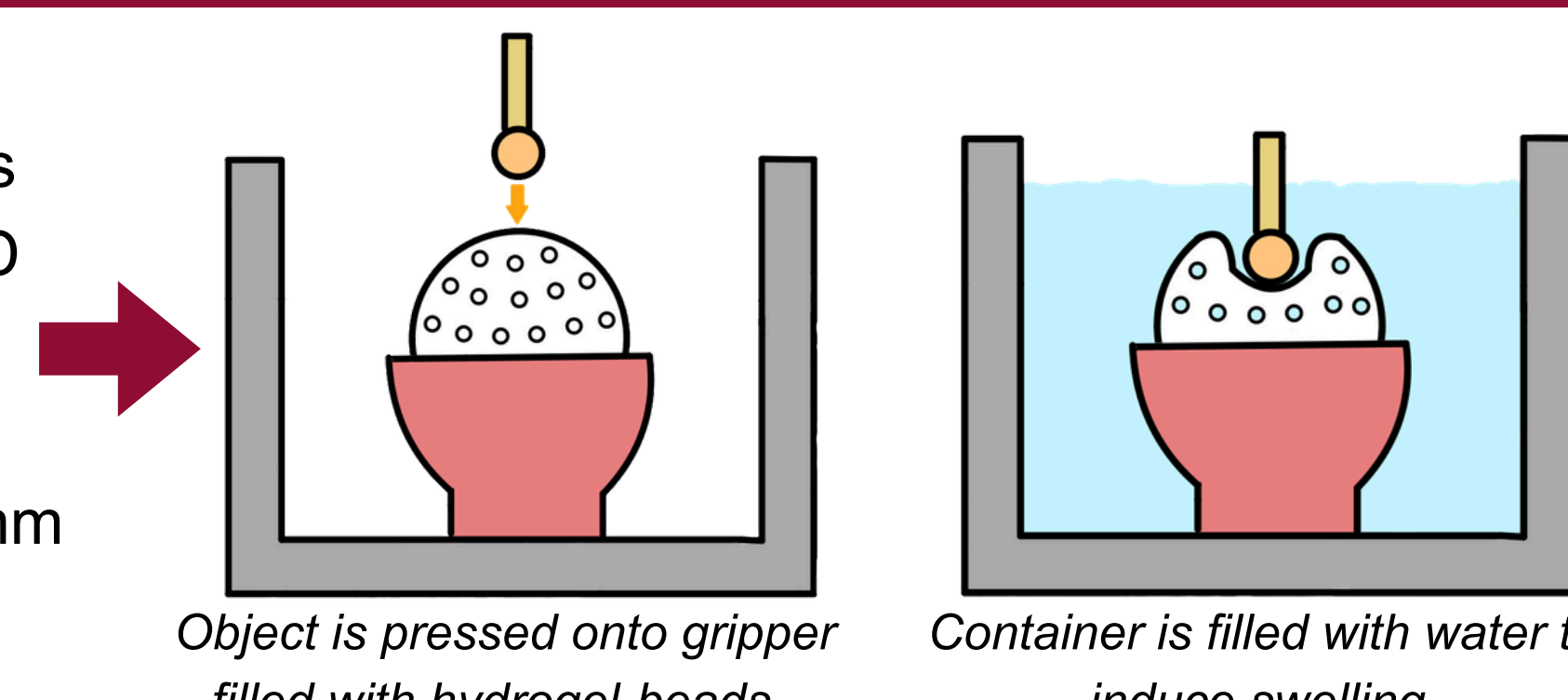


Hydrogels are 3D polymer networks made from crosslinked polymer chains. When swelling is induced, the 3D network expands (hydrogel's water content increases).

Testing Set-Up:



- Experimental set-up was designed in CAD and 3D printed in PLA.
 - Container:
 - 120 mm x 120 mm x 83 mm



Object is pressed onto gripper filled with hydrogel-beads. Container is filled with water to induce swelling.

Results

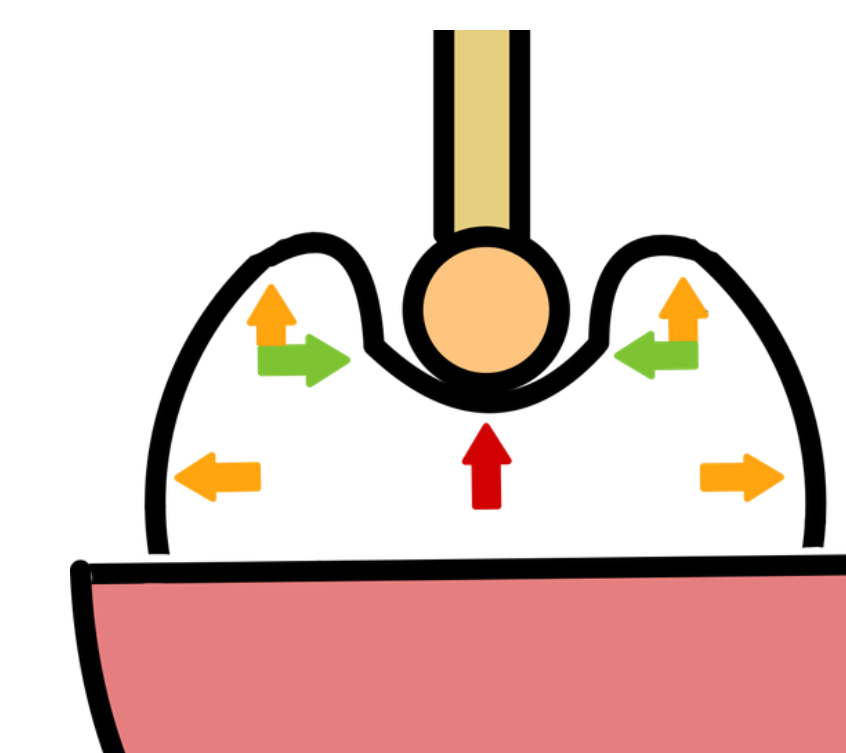


Gripper containing 10 grams of hydrogel beads results after swelling for 5 minutes.

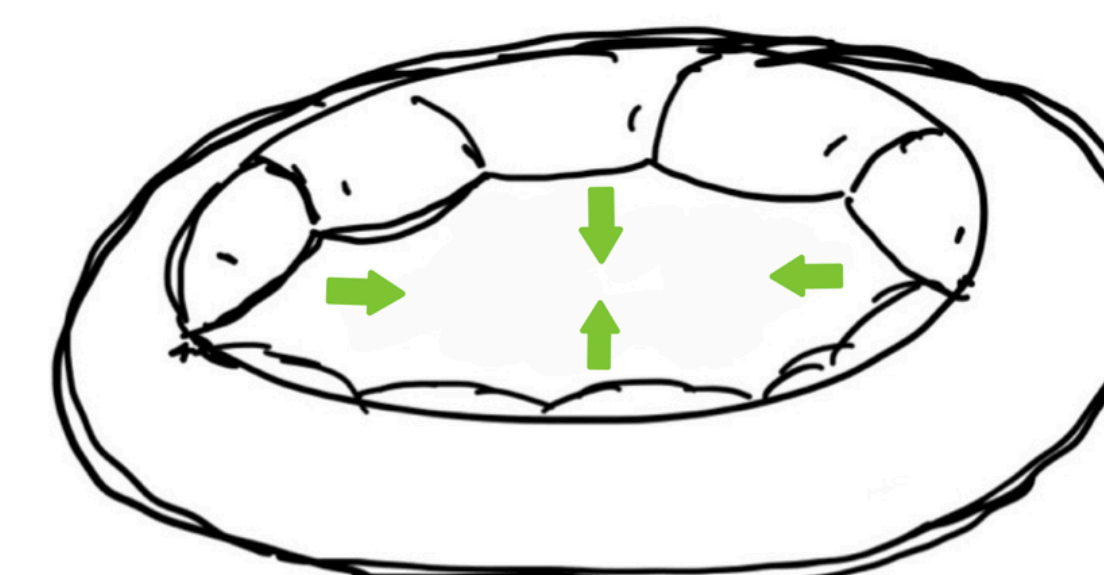
Observations:

- Object was not grasped** but immediately pushed back up.
- Despite a good indication of membrane flexibility, the **desired envelopment of the object had not been reached**.
- Good indication of **high internal pressures** being generated upon swelling.

Discussion



Green - wanted forces, Orange - neutral forces, Red- unwanted forces



Although not universal, this O-Ring gripper design demonstrates the desired forces we want to mimic.

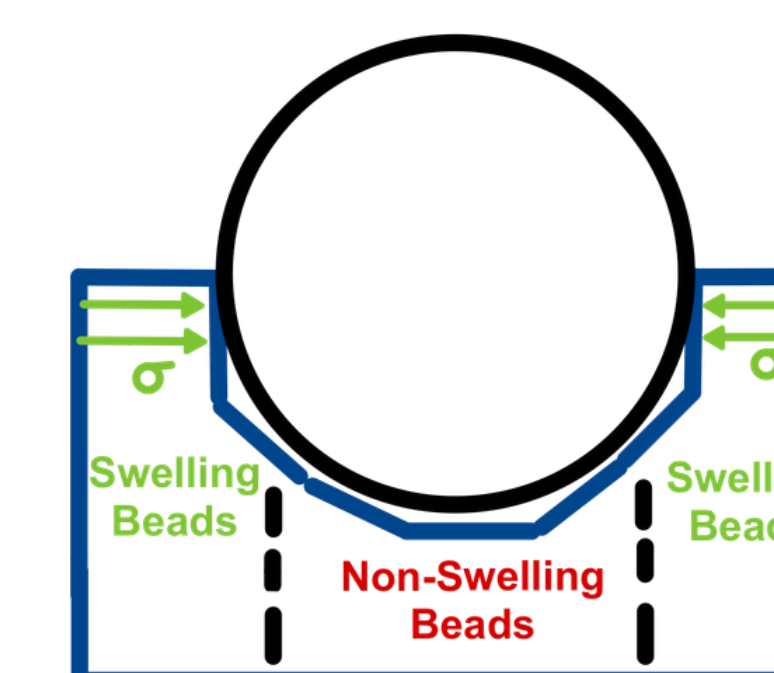
Hydrogel Bead Swelling:

- Due to the membrane stretching isotropically, **unwanted vertical forces** acting against the object are created.
 - Need to maximize compressive forces relative to vertical forces.**

Prioritization of compression forces for grasping by:

- Gripper redesign.**
- Strategically incorporating **swelling and non-swelling beads** or beads that swell less.

Next Steps

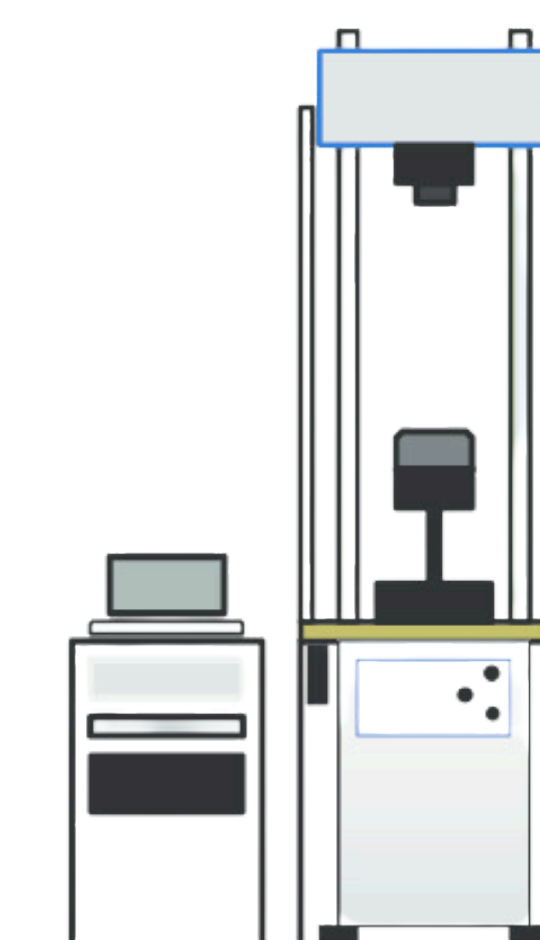


Next Gripper Iteration:

- Experimenting with **swelling and non-swelling hydrogel beads**.
- Optimize gripper shape** for greater envelopment.

Quantitative Testing:

- Measure gripping strength before slippage on the Instron Machine.
- Important Variables:**
 - Membrane Porosity:**
 - number of holes/surface area
 - Membrane thickness**
 - Object contact angle:**
 - arc length of contact/radius of object
 - Hydrogel-bead quantity**



References

- Brown, E., Rodenberg, N., Amend, J., Jaeger, H. M., & Meiron, D. (2010). Universal robotic gripper based on the jamming of granular material. *Proceedings of the National Academy of Sciences*, 107(44), 18809-18814. <https://doi.org/10.1073/pnas.1003250107>
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